

AirSense

BIC Islamabad x BIC Karachi

An IoT and Artificial Intelligence Initiative for Real-Time Air Quality Monitoring, Intelligent Forecasting and Operational Decision Support

Developed during COIL AI | A Proposal for Budget Approval and Pilot Deployment

SUBMITTED TO

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PKR 20,000

One-Time Investment

**2 BIC
Campuses**

Islamabad + Karachi

**Live Public
Dashboard**

BIC-Branded Online

COIL AI

Internationally Recognised

Origin

1. Executive Summary

We are not here to propose an experiment. We are here to ask for the resources to deploy something we have already built.

AirSense is a fully developed, AI-powered air quality intelligence platform, built by BIC students during the COIL AI international academic programme. The system is not a concept or a prototype. It has an architecture, a machine learning pipeline with trained models, a working dashboard, and a validated hardware specification. What it does not yet have is real, locally collected campus data to make its forecasts accurate for the specific environment in which it will run.

This proposal requests PKR 20,000 to fund the installation of a rooftop sensor unit at BIC Islamabad, with a parallel deployment coordinated at BIC Karachi. Over three months, both sensors will collect continuous air quality and meteorological data. That data will retrain our existing machine learning models on real BIC conditions, and the results will be displayed on a live, publicly accessible dashboard that carries the Beaconhouse International College name.

BIC's investment of PKR 20,000 will produce a working AI system deployed on two campuses, a live public dashboard showcasing BIC student innovation, and trained models calibrated to BIC's own environment. The institute does not just fund the project. The institute owns the outcome.

We believe this project will succeed because we have already done the hard part. The models are built. The dashboard is built. The hardware is specified and sourced. We are asking for one-time funding to turn a completed system into a deployed one.

2. Born from COIL AI

2.1 An Internationally Recognised Academic Initiative

AirSense was conceived, developed, and refined during the COIL AI programme, a structured international academic collaboration that brought together students and institutions across multiple countries to build technology solutions with real-world impact. The project was developed under formal academic oversight and competitive evaluation, and was recognised for the quality of its design and the ambition of its scope.

COIL AI was not a classroom exercise. It was a competitive innovation environment in which student teams were expected to produce systems that could survive scrutiny from academic and professional evaluators. AirSense emerged from that environment as a platform with a clear product identity, a defined market position, and a technical architecture capable of real-world deployment.

The fact that AirSense was developed under the COIL AI framework gives BIC's investment a layer of external validation that most student projects do not have. This is not an idea developed in a lab. It is a system developed through an internationally structured programme and evaluated against international standards.

2.2 Why IoT Implementation Matters

AirSense is, at its core, an IoT-based system. It connects physical hardware deployed in a real environment to a cloud-connected data pipeline and an AI-powered analytics layer. This is not incidental to the project. It is the reason the project has value.

IoT-based implementation bridges the gap between academic theory and practical engineering in a way that no simulation or dataset analysis can replicate. Students who design, assemble, deploy, and operate a real hardware system in a real outdoor environment are acquiring skills that industry actively demands and educational institutions rarely deliver. BIC, by hosting this deployment, positions itself as an institution that does not just teach AI and computer science but actively enables its students to build and deploy production-grade systems.

The rooftop sensor at BIC Islamabad is not a piece of student equipment. It is a functioning node of an AI-connected environmental intelligence network, operated by BIC students, on BIC infrastructure, producing data that will be publicly available and institutionally attributed. That is the practical difference between theoretical education and applied innovation.

2.3 The Problem We Are Solving

Air quality in Pakistan's urban centres is a genuine and worsening public health concern. PM2.5, fine particulate matter smaller than 2.5 micrometres, is the primary driver of air pollution-related health impacts, contributing to respiratory illness, cardiovascular disease, and reduced cognitive performance at sustained concentrations. Islamabad and Karachi both experience measurable air quality deterioration driven by traffic, construction, and seasonal factors.

Institutions currently make decisions about outdoor activities, ventilation, and physical education without access to real-time, locally relevant environmental data. There is no live, publicly available air quality reading from the BIC campus today. AirSense will change that, and BIC will be the institution that made it happen.

3. The AirSense Platform

3.1 What It Does

AirSense does not display data. It converts data into decisions. This is the core distinction between AirSense and a conventional air quality monitor. A standard monitor tells you that PM2.5 is at a certain level. AirSense tells you what to do about it in the next hour, the next three hours, and the next day.

The system operates across four connected layers, from physical measurement through to plain-language operational guidance.

Layer	Technology	What It Delivers
1. IoT Sensing	PMS7003 Laser Sensor + BME280 + Rain Sensor + ESP32	Real-time measurement of PM1, PM2.5, PM10, temperature, humidity, barometric pressure, and rainfall events from the rooftop. Data logged locally and transmitted via Wi-Fi to the processing layer.
2. Data Pipeline	MicroSD Logging + Campus Wi-Fi + Public Weather API	Continuous data collection with dual-channel redundancy (local SD + Wi-Fi transmission). Public weather feed supplies wind speed and direction for dispersion modelling.
3. AI Forecasting	In-House ML Models (XGBoost, Random Forest, LightGBM)	Trained on BIC campus data to predict PM2.5 concentrations 1 hour, 3 hours, 6 hours, and 24 hours ahead. Walk-forward validation ensures forecasts reflect real operational performance.
4. Decision Layer	Operational Playbook + AI Language Model Advisor	Translates forecast outputs into plain-language operational recommendations. The advisor responds to natural-language queries with forecast-informed guidance specific to current and predicted conditions.

3.2 In-House Machine Learning Models

A critical feature of AirSense that distinguishes it from commercial air quality services is that every model powering its forecasts was built in-house by BIC students. These are not licensed algorithms or third-party tools. They are original implementations developed, tested, and refined during the COIL AI programme.

Model	Technique	Role in AirSense
AQI Predictor	XGBoost Gradient Boosting	Primary forecasting model. Trained on lag features, meteorological variables, and temporal patterns to predict PM2.5 concentrations at each forecast horizon.
Multi-Step Forecaster	Gradient Boosting Regressor	Produces sequential 1h-to-24h forecast chains. Feeds the 72-hour visualisation in the dashboard.
Health Risk Classifier	Random Forest Classifier	Maps forecast PM2.5 levels to health risk tiers for different user groups including students, staff, and outdoor workers.
Anomaly Detector	Isolation Forest	Identifies abnormal air quality events for alert generation and data quality control.

These models were developed and tested on publicly available datasets during COIL AI. The campus pilot retrains them on locally collected BIC data, producing campus-specific versions calibrated to the actual conditions of this environment. The result is a set of AI models that are proprietary to BIC, trained on BIC data, and optimised for BIC's specific campus context.

No external vendor is providing the AI. BIC students built it. The campus pilot makes it accurate for BIC specifically. The intellectual property belongs to the institution.

4. Campus Deployment Plan

4.1 What We Will Install and How

The hardware assembly is a single, weatherproofed sensor unit mounted on the BIC Islamabad rooftop. It is compact, professionally assembled, and draws from a standard power outlet. Once installed and verified, it requires no daily attention. The following table outlines the deployment from procurement to live operation.

Phase	Timeframe	What Happens
0	Days 1-7	Hardware Procurement - Place orders with verified local vendors: Embeded Studio (PM sensor), Digilog.pk (ESP32), Electrobes (BME280, anemometer), A.E Solution (enclosure), Daraz.pk (storage, wiring), Clopal Online (extension lead). - Confirm stock, delivery timelines, and specifications before payment. - Estimated total cost: PKR 17,787 with PKR 2,213 held as contingency buffer.
1	Days 6-9	Bench Assembly and Validation - Assemble components on workbench: wire sensors to ESP32, connect SD logging module, flash firmware. - Test sensor readings, SD logging, and Wi-Fi transmission. - No component proceeds to rooftop installation until bench validation passes.
2	Day 9-10	Rooftop Installation - Mount weatherproof enclosure on BIC rooftop using stainless clamps. - Position Wi-Fi extender for rooftop coverage and seal power lead. - End-to-end telemetry check. System goes live. Pilot clock starts.
3	Months 1-3	Autonomous Data Collection - Continuous 24/7 sensing every 5-10 minutes with zero daily maintenance. - Operates autonomously through academic breaks to establish baseline. - Dual-channel redundancy (SD card + Wi-Fi sync) guarantees zero data loss.

4	Months 2-3	Model Retraining and Validation • Clean and quality-control the dataset. • Retrain in-house ML models on BIC campus data. • Validate predictions across 1h, 3h, 6h, and 24h horizons.
5	End Month 3	Reporting and Dashboard Launch • Submit comprehensive pilot completion report to BIC management. • Launch live public dashboard with BIC campus data. • Archive dataset and trained models for institutional ownership.

5. A Collaboration Between Two BIC Campuses

5.1 One Project, Two Cities, One Network

AirSense is not a single-campus initiative. From day one, both Beaconhouse International College Islamabad and Beaconhouse International College Karachi are participating in the project as equal partners. The same hardware specification, the same data pipeline, and the same machine learning models will be deployed at both institutions simultaneously, under the shared coordination of Project Lead Munim Qureshi.

This inter-campus structure is strategically significant. It transforms AirSense from a local student project into a BIC network initiative: two BIC campuses, in two different cities, collaborating on a jointly developed AI and IoT platform.

BIC Islamabad (Flagship Site)	BIC Karachi (Coastal Sister Site)
Supervisor: Ms. Sahifa Alam Head of CSSE / AI Department • Primary deployment site and ML training anchor. • Full development team on-site for assembly and monitoring. • Inland microclimate: cold winter smog, temperature inversions.	Supervisor: Mr. Sajid Head of CSSE / AI Department • Parallel deployment coordinated by Munim Qureshi. • Identical hardware stack and data pipeline. • Coastal climate: high humidity, sea breeze dispersion.

5.2 How Both Campuses Develop the System Together

The collaboration between BIC Islamabad and BIC Karachi is not a division of labour. It is a shared development process. Both campuses contribute data to the same shared pipeline. Both campuses benefit from models that are trained on a combined dataset representing Pakistan's two most important urban air quality environments.

No other student initiative in the BIC network currently involves two campuses actively developing and deploying the same AI system in parallel. This is not a student project that happens to have two locations. It is a BIC innovation network in its founding phase.

6. How AirSense Benefits BIC

6.1 A Live, BIC-Branded Public Dashboard

Upon completion of the pilot and verification of model accuracy, the AirSense dashboard will be made publicly accessible online. Any person visiting the platform will see live air quality data from BIC campus sensors, with BIC Islamabad and BIC Karachi identified as the host institutions.

The dashboard is not a report or a presentation. It is a permanent, publicly accessible demonstration of BIC student capability, operating in real time and freely available to anyone who looks for it.

6.2 Campus Benefits Summary

Benefit Area	What It Means for BIC
Live Campus Air Quality Data	BIC Islamabad and Karachi receive real-time, locally measured air quality readings. Enables data-driven decisions about outdoor activities, PE schedules, and ventilation.
Publicly Available Dashboard	A live, BIC-branded platform displaying real-time campus data and AI forecasts. Converts a one-time PKR 20k purchase into ongoing institutional visibility.
Proprietary AI Models	Four in-house ML models, developed by BIC students and retrained on BIC campus data, become institutional assets owned by BIC. Zero external vendor dependency.
COIL AI Credential	Origin in a structured international academic programme gives BIC a distinct credential: a student-built, internationally validated AI and IoT platform deployed on campus.
Applied AI & Curriculum	CSSE and AI departments gain a live case study in applied IoT deployment and ML that future cohorts can study and extend.
Institutional Research Data	Three months of continuously collected environmental data from both campuses supporting student research projects and academic publications.
Inter-Campus Innovation	Establishes a working model for cross-city technical collaboration across BIC campuses nationwide.

7. Dashboard Overview

The AirSense dashboard is the public face of the platform. It is the interface through which campus data, AI forecasts, and operational recommendations are presented. Below is an overview of its key panels.

7.1 Live AQI and Station Overview

Displays current AQI readings, PM2.5 concentrations with status indicators, and active station status for BIC Islamabad and BIC Karachi.

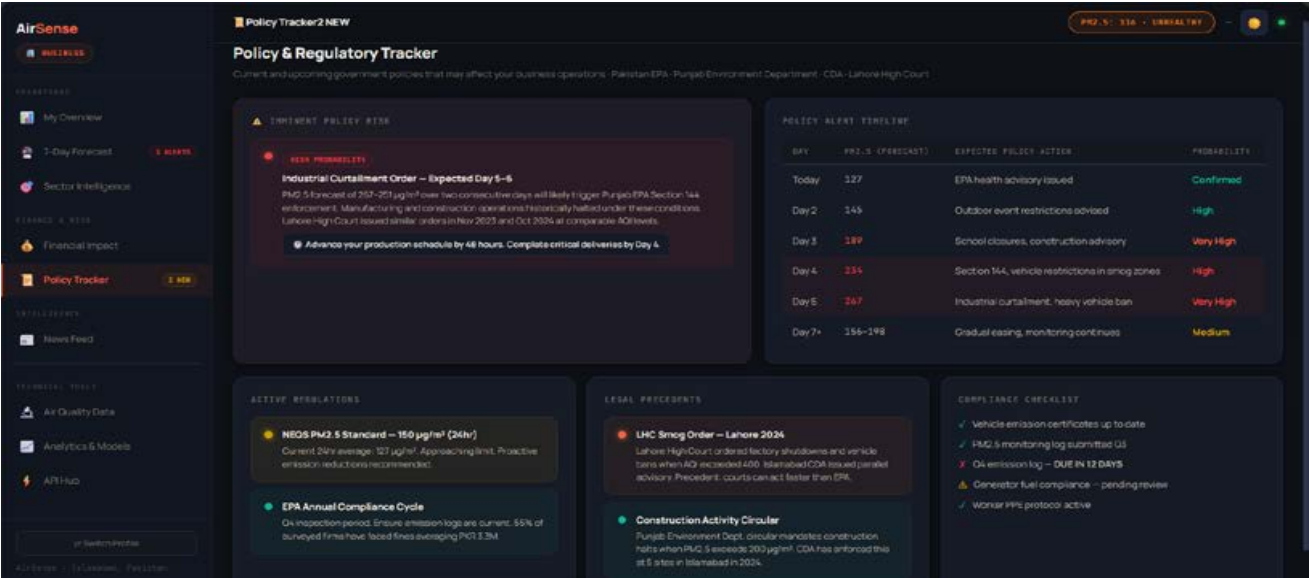


Figure 7.1a: AirSense Executive Overview Panel displaying real-time PM2.5 and predictive risk trends.

7.2 AI Forecast Panel and Policy Tracker

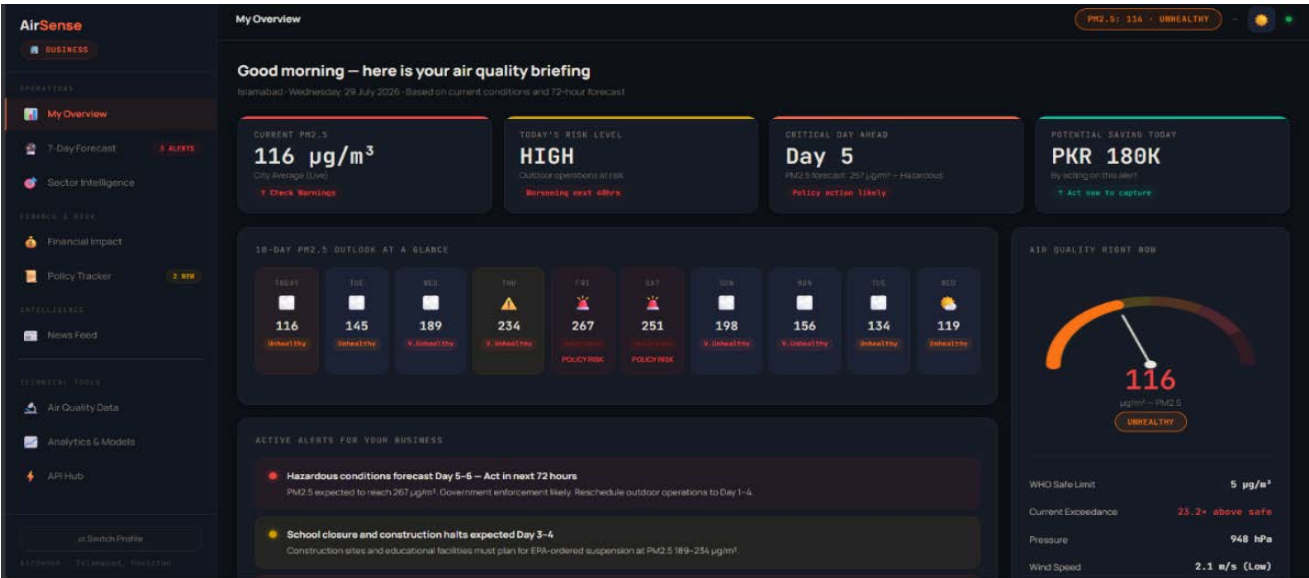


Figure 7.1b: Policy & Regulatory Tracker tracking environmental directives and compliance.

7.3 AI Decision Advisor

The Decision Advisor accepts natural-language queries and responds with forecast-informed operational guidance tailored to campus needs.



Figure 7.3: AI Decision Advisor converting forecasts into administrative guidance.

7.4 Health Risk Panel and Dispatch Calculator

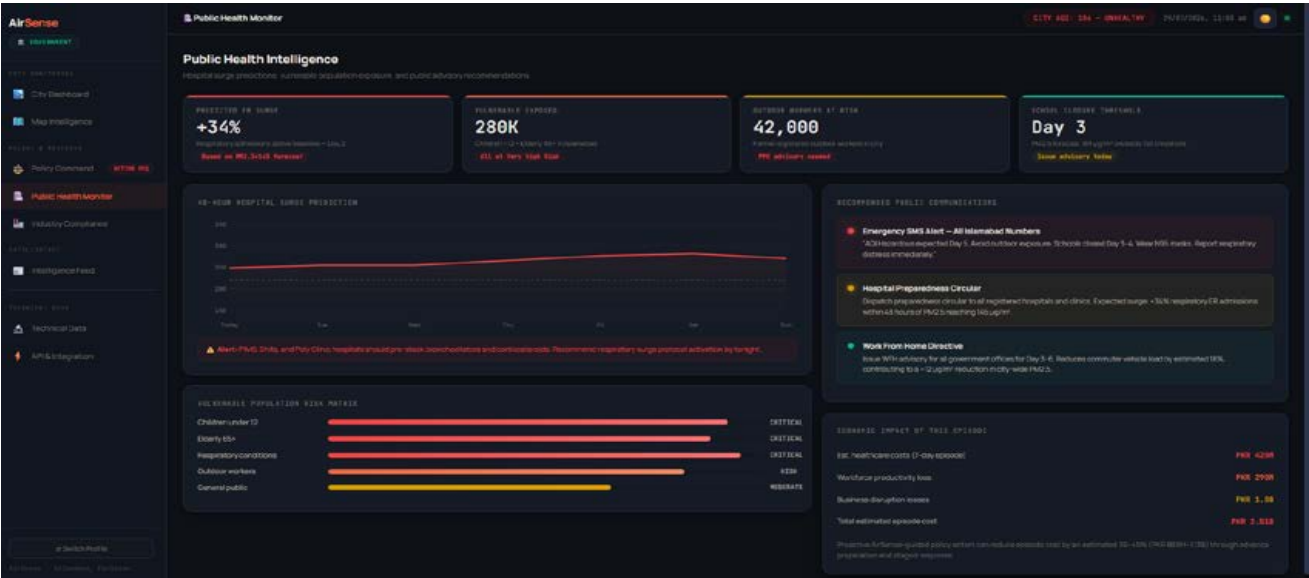


Figure 7.4: Health Risk Panel calculating safe dispatch windows for outdoor activities.

8. Operational Decision Making and Future Vision

8.1 The Real Value of AirSense: Decisions, Not Data

Air quality data without interpretation is noise. What institutions need is not a number on a screen, but clear guidance on what that number means for operations in the coming hours.

Sector	Trigger Condition	Operational Action Recommended
Schools & Campuses	PM2.5 rising in next 3 to 6 hours	Reschedule outdoor sports. Increase indoor filtration. Alert physical education staff. Adjust fresh-air ventilation intake.
Logistics & Delivery	High PM2.5 morning peak forecast	Shift loading and route dispatch to lower exposure windows. Alert drivers. Reschedule high-exertion tasks.
Construction Sites	High PM2.5 + low wind dispersion	Reduce dust-generating activity. Increase wet suppression. Pause outdoor work if health thresholds are exceeded.
Retail & Hospitality	Poor air quality episode forecast	Adjust HVAC settings. Display indoor air quality messaging to customers. Protect outdoor staff.
Corporate & ESG	Repeated threshold exceedance	Generate automated alert logs for incident records, mitigation evidence, and corporate sustainability reporting.

8.2 How AirSense Can Be Offered to the Market

The campus pilot is the first step toward a commercially viable product offered as a SaaS subscription service to logistics operators, construction firms, educational institutions, and corporate campuses across Pakistan.

8.3 From BIC Pilot to BIC-Owned Platform

The most important long-term benefit is institutional ownership of a validated AI platform. BIC is established as the founding home of the technology.

9. Phase Roadmap

Phase	Stage	Description	Key Output for BIC
1	BIC Campus Pilot	Deploy rooftop sensors at BIC Islamabad & Karachi. Collect 3 months data. Retrain ML models on BIC campus data.	Validated dataset, BIC-calibrated AI models, live public dashboard, completion report.
2	Model Refinement	Retrain models on full dataset. Improve feature engineering. Expand to additional Beaconhouse network campuses.	Higher-accuracy models, multi-campus BIC dataset, proven platform.
3	Commercial Launch	Deploy SaaS subscription platform targeting logistics, construction, retail, and educational clients.	Revenue-generating SaaS product; BIC IP partnership.
4	National Scale	Expand to additional cities and sectors across Pakistan.	BIC positioned as founding institution of national environmental AI network.

10. Budget Request: PKR 20,000

10.1 What This Investment Covers

The PKR 20,000 requested represents a single, one-time capital expenditure. There are no recurring costs or subscription fees during the pilot period.

#	Component	Vendor	Qty	Cost (PKR)
1	PMS7003 Laser Dust Sensor	Embedded Studio	1	4,200
2	ESP32 WROOM-32D Board	Digilog.pk	1	1,160
3	BME280 Temp/Humidity/Pressure Sensor	Electrobes	1	850
4	Arduino MicroSD Module	Electrobes	1	150
5	Samsung 16GB MicroSD Card	Daraz.pk	1	1,450
6	IP65 Waterproof Enclosure	A.E Solution	1	750
7	Jumper Wire Kit, 120 pcs	Daraz.pk	1	440
8	5V 2A AC/DC Power Adapter	Electronics Hub	1	170
9	Cable Ties, 100-pack	Daraz.pk	1	120
10	Stainless Hose Clamps (2 pcs)	DreamsMart.pk	1	60
11	Raindrop Sensor Module A	Digilog.pk	1	160
12	Rain Drop Moisture Sensor Module B	Electrobes	1	150
13	UNI-T UT363 Digital Anemometer	Electrobes	1	4,250
14	Clopal 10m Heavy-Duty Lead	Clopal Online	1	3,295
15	GMSA RTV Weatherproof Sealant	Expert Tools	1	390
16	Mounting Nuts & Bolts (8 pcs)	Multan Elec.	8	192
17	Radiation Shield (Local Sourcing)	Local Supplier	1	TBD
	Hardware Subtotal (Items 1-16)			PKR 17,787
	Radiation Shield + Contingency Buffer			PKR 2,213
	TOTAL REQUESTED BUDGET			PKR 20,000

11. Deliverables at Pilot Completion

No.	Deliverable	Description
1	BIC Campus Air Quality Dataset	Three months of continuous, quality-controlled hourly records from BIC Islamabad and Karachi. Full institutional ownership.
2	Retrained In-House ML Models	Four in-house models retrained on BIC campus data with performance metrics. Models remain institutional assets.
3	Live Public Dashboard	Publicly accessible online platform displaying live BIC campus data, providing permanent institutional visibility.
4	Decision Framework Evaluation	Structured assessment of operational decision layer accuracy and utility under real campus conditions.
5	Pilot Completion Report	Comprehensive technical and strategic report submitted to BIC management and departmental supervisors.

12. Why BIC Should Say Yes

We are asking for PKR 20,000. We are offering BIC a publicly accessible, BIC-branded AI platform displaying live data from campus sensors, four in-house machine learning models retrained on BIC data, and a three-month environmental dataset from two BIC campuses.

We believe AirSense will succeed.

We have built the system. We have built the team. We have the academic backing.

All we need is the opportunity to run it on real BIC data from a real BIC rooftop.

Give us PKR 20,000 and three months. We will give BIC something to be proud of.

The team respectfully requests approval of the following:

1	Approval of a one-time hardware budget of PKR 20,000 for procurement of sensor components.
2	Permission to install the weatherproofed sensor assembly on the BIC Islamabad rooftop for three months.
3	Facilitation of access to a rooftop power point and setup of a campus Wi-Fi extender.
4	Official recognition of AirSense as a supported BIC initiative, granting access to computing resources for model training.

Respectfully submitted,

Munim Qureshi (Project Lead | AirSense | BIC Islamabad and BIC Karachi)

Hiba Safdar | M. Abbas | M. Bilal Ahmed | Komal Alishba (AirSense Development Team)

Approval Sign-Off

Role	Name	Signature	Date
Head of Institute	Ms. Saba Ahson		
Head of CSSE/AI, BIC Islamabad	Ms. Sahifa Alam		
Head of CSSE/AI, BIC Karachi	Mr. Sajid		
Project Lead	Munim Qureshi	<i>Munim</i>	29/07/2026